

Ayomide Abilewa

Electronic and Electrical Engineering student, focused on embedded systems and instrumentation

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PROFILE

Electronic and electrical engineering student who builds embedded and computer-vision systems, with a habit of designing for the conditions that break them: poor lighting, no network, unreliable input. Currently rotating through the electrical and instrumentation team at Chevron Nigeria, working with the control loops and signal standards behind the same measurement problems at industrial scale.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, JavaScript, Verilog/HDL, Bash

Embedded & Hardware: Arduino, ESP32, STM32, Raspberry Pi, FPGA, I2C / SPI / UART, Sensor integration

AI & Machine Learning: YOLOv8/v11, OpenCV, TensorFlow (fundamentals), PyTorch (fundamentals), Weighted Boxes Fusion

Instrumentation & Control: P&ID interpretation, PID control loops, Pneumatic systems, I/P converters, 4–20 mA and 3–15 PSI standards

Electrical Systems: SLD interpretation, Circuit breaker sizing, ESP systems, Power transmission, Arc flash safety, Hazardous area classification

Tools: VS Code, Git/GitHub, MATLAB/Simulink, KiCad, Flask, React/TypeScript, PostgreSQL, Blynk IoT

ENGINEERING EXPERIENCE

Facilities Engineering Management Intern (SIWES) · Chevron Nigeria Limited

Mar 2025–Present

Lagos, Nigeria

- Rotate across the Electrical and Instrumentation (E&I) team, working on oil and gas field instrumentation and electrical systems.
- Interpret P&IDs and study control loop configurations, pneumatic signal standards (3–15 PSI), I/P converters and hazardous area classification.
- Study ESP systems, power transmission, single line diagram (SLD) interpretation and circuit breaker specifications.
- Apply electrical safety practice including arc flash awareness, bonding and grounding.

Research and Development Intern (SIWES) · ACE Quanser Robotics Lab, Obafemi Awolowo University

2024

Ile-Ife, Nigeria

- Operated the QDrone quadrotor UAV, QCar autonomous ground vehicle and QArm 6-DOF manipulator using MATLAB and Simulink.
- Applied PID control, sensor feedback and real-time system analysis to live hardware rather than simulation alone.

Embedded Systems and EEE Tutor · Astar Tutorials

2023–Present

Ile-Ife, Nigeria

- Teach circuit analysis, digital logic and microcontroller programming to engineering students.

SELECTED PROJECTS

aniwe · Python, Streamlit, OpenRouter, Vision-language models

Aug 2026

- Built an AI costume-rating booth in Python and Streamlit with ten original hosts, per-session mood and quirk variation, and vision-language scoring across seven criteria collapsed into a single ranked score.
- Designed three-tier text-to-speech degradation (ElevenLabs, offline Piper, browser speech) and local faster-whisper speech-to-text so a blocked network reduces quality without stopping the system.

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Wrote 214 offline tests running in about 1.5 seconds with no network, camera or model downloads, covering truncated-JSON repair, persona rotation, rank boundaries and the network fallback chain.

github.com/ayomide-abilewa/aniwe

AI-Powered Smart Attendance System · Python, OpenCV, YuNet, SFace

Feb – Mar 2026

- Built an AI-powered attendance system using YuNet face detection and SFace recognition with cosine similarity matching, delivering the computer-vision pipeline and a Flask REST API backend.
- Developed the React, TypeScript and Tailwind frontend and a PostgreSQL schema covering students, courses, enrolment, sessions and attendance records.
- Defended the system at a final-year group project review and was invited to develop an extended production version.

github.com/ayomide-abilewa/Smart-Attendance-System

Live Video Pipe Anomaly Detection · Python, YOLOv8, YOLOv11, Weighted Boxes Fusion

2026 – present

- Developing a two-stage cascaded detection architecture for real-time pipe anomaly identification, targeting industrial inspection.
- Built a parallel ensemble of RGB and edge-enhanced YOLO models merged with Weighted Boxes Fusion to improve precision across varied lighting and surface conditions.

Multispectral Acquisition Device for Hydroponics · ESP32, GY-2561, GY-31, I2C

2025

- Built a multi-sensor ESP32 acquisition node measuring light intensity and colour spectrum with the GY-2561 and GY-31, streaming live to a Blynk IoT Cloud dashboard.
- Integrated multiple I2C sensors on a single ESP32, handling bus arbitration and sensor calibration.

Smart Temperature-Controlled Fan · Arduino, C++, PWM, NTC thermistor

2024

- Led a team building a closed-loop fan speed controller on Arduino, driving PWM output from real-time NTC thermistor feedback.

EDUCATION

B.Sc. Electronic and Electrical Engineering · Obafemi Awolowo University

2021–2027 (expected)

Ile-Ife, Osun State, Nigeria

TRAINING AND CERTIFICATIONS

- Operational Excellence Training. Chevron, 2025.
- Cyber Security and Data Privacy Awareness. Chevron, 2025.
- FPGA Design and HDL/Verilog Workshop (SWEP). Faculty of Engineering, Obafemi Awolowo University, 2024.
- Robotics Lab Orientation and Operations. ACE Quanser Lab, Obafemi Awolowo University, 2024.

LEADERSHIP AND TEACHING

SPAW Project Lead and Syllabus Team Lead · IEEE OAU Student Branch

2024–Present

- Lead SPAW 3.0, a six-session embedded systems and entrepreneurship curriculum for secondary school students.